Class 11: AlphaFold

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```r
library(bio3d)

pdb <- read.pdb("hivpr_dimer_23119/hivpr_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v

pdb
```

```
 Call:  read.pdb(file = "hivpr_dimer_23119/hivpr_dimer_23119_unrelaxed_rank_001_alphafold2_mu

   Total Models#: 1
     Total Atoms#: 1514,  XYZs#: 4542  Chains#: 2  (values: A B)

     Protein Atoms#: 1514  (residues/Calpha atoms#: 198)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 0  (residues: 0)
     Non-protein/nucleic resid values: [ none ]

   Protein sequence:
      PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
      QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
      ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
      VNIIGRNLLTQIGCTLNF

+ attr: atom, xyz, calpha, call
```

Make a vector of input PDB file names that we can read into R

```r
pdbfiles <- list.files("hivpr_dimer_23119/",
                       pattern=".pdb",
                       full.names = T)
```

```r
library(bio3d)

# Read all data from Models
#  and superpose/fit coords
pdbs <- pdbaln(pdbfiles, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
hivpr_dimer_23119/hivpr_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_00
hivpr_dimer_23119/hivpr_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_00
hivpr_dimer_23119/hivpr_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_00
hivpr_dimer_23119/hivpr_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_00
hivpr_dimer_23119/hivpr_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_00
.....

Extracting sequences

pdb/seq: 1   name: hivpr_dimer_23119/hivpr_dimer_23119_unrelaxed_rank_001_alphafold2_multimer
pdb/seq: 2   name: hivpr_dimer_23119/hivpr_dimer_23119_unrelaxed_rank_002_alphafold2_multimer
pdb/seq: 3   name: hivpr_dimer_23119/hivpr_dimer_23119_unrelaxed_rank_003_alphafold2_multimer
pdb/seq: 4   name: hivpr_dimer_23119/hivpr_dimer_23119_unrelaxed_rank_004_alphafold2_multimer
pdb/seq: 5   name: hivpr_dimer_23119/hivpr_dimer_23119_unrelaxed_rank_005_alphafold2_multimer
```

```r
pdbs
```

```
                               1        .         .         .         .         50
[Truncated_Name:1]hivpr_dime   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:2]hivpr_dime   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:3]hivpr_dime   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:4]hivpr_dime   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:5]hivpr_dime   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
                               **************************************************
                               1        .         .         .         .         50

                              51        .         .         .         .         100
[Truncated_Name:1]hivpr_dime   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:2]hivpr_dime   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:3]hivpr_dime   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:4]hivpr_dime   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:5]hivpr_dime   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
                               **************************************************
                              51        .         .         .         .         100
```

```
                               101        .         .         .         .         150
[Truncated_Name:1]hivpr_dime   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:2]hivpr_dime   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:3]hivpr_dime   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:4]hivpr_dime   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:5]hivpr_dime   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
                                **************************************************
                               101        .         .         .         .         150

                               151        .         .         .         .       198
[Truncated_Name:1]hivpr_dime   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]hivpr_dime   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]hivpr_dime   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:4]hivpr_dime   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]hivpr_dime   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
                                ************************************************
                               151        .         .         .         .       198

Call:
  pdbaln(files = pdbfiles, fit = TRUE, exefile = "msa")

Class:
  pdbs, fasta

Alignment dimensions:
  5 sequence rows; 198 position columns (198 non-gap, 0 gap)

+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

Using the **rmsd()** function to calculate the RMSD between all pairs models

```r
rd <- rmsd(pdbs, fit=T)
```

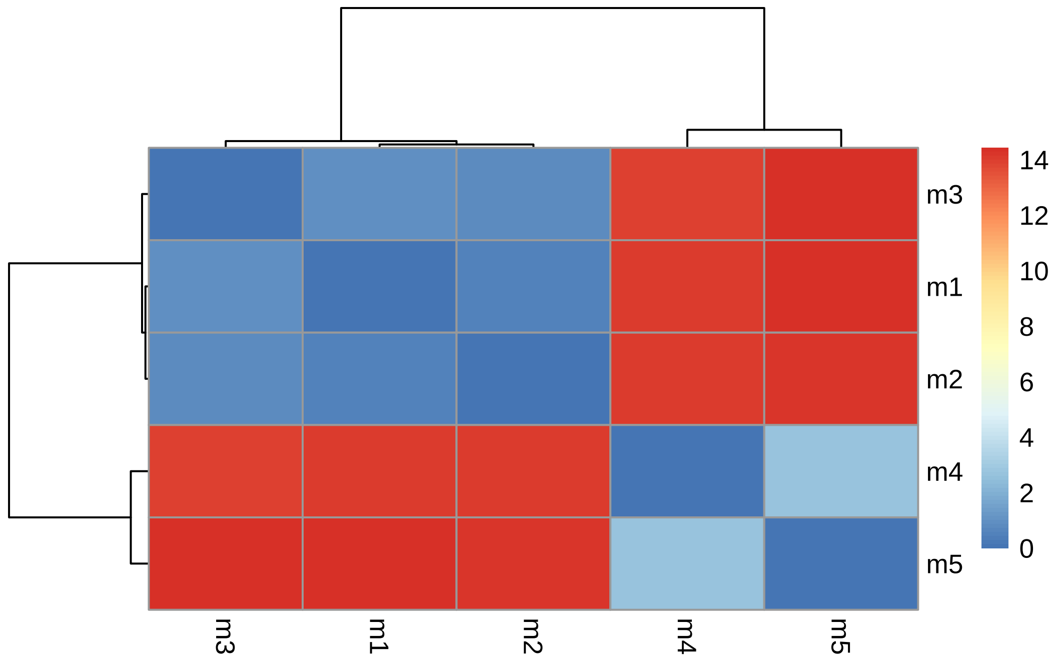
```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 198 non NA positions
```

```r
range(rd)
```

```
[1]  0.000 14.428
```

Drawing a heatmap of the RMSD matrix values

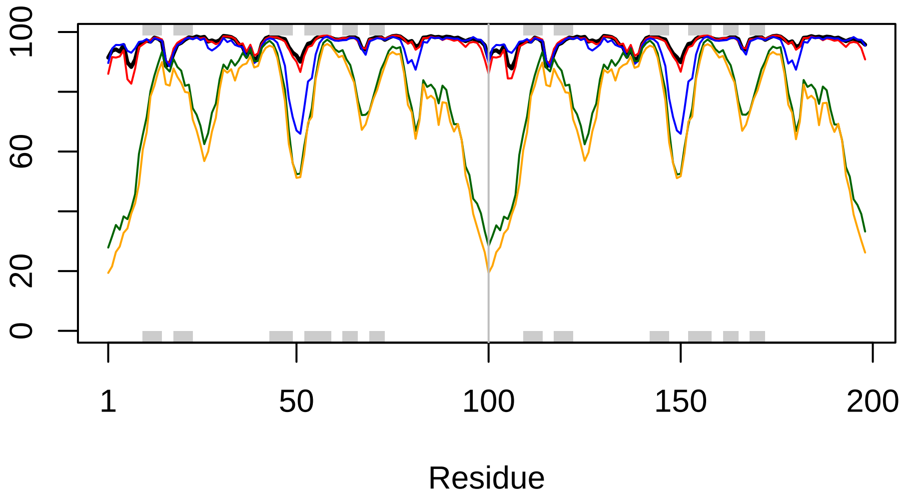
```r
library(pheatmap)

colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```




```r
# Read a reference PDB structure
pdb <- read.pdb("1hsg")
```

```
## Note: Accessing on-line PDB file
```

```r
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdb)
points(pdbs$b[2,], typ="l", col="red")
points(pdbs$b[3,], typ="l", col="blue")
points(pdbs$b[4,], typ="l", col="darkgreen")
points(pdbs$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```



The superposition/fitting can be improved by finding the most consistent "rigid core" that is common across all the models.

```r
core <- core.find(pdbs)
```

```
 core size 197 of 198  vol = 8545.023
 core size 196 of 198  vol = 7893.807
 core size 195 of 198  vol = 3573.721
 core size 194 of 198  vol = 1850.924
 core size 193 of 198  vol = 1697.206
 core size 192 of 198  vol = 1612.782
 core size 191 of 198  vol = 1530.216
 core size 190 of 198  vol = 1447.416
 core size 189 of 198  vol = 1377.126
 core size 188 of 198  vol = 1303.836
 core size 187 of 198  vol = 1239.049
 core size 186 of 198  vol = 1188.148
 core size 185 of 198  vol = 1118.483
 core size 184 of 198  vol = 1071.683
 core size 183 of 198  vol = 1034.024
 core size 182 of 198  vol = 980.844
 core size 181 of 198  vol = 942.256
 core size 180 of 198  vol = 911.403
```

```
core size 179 of 198  vol = 879.764
core size 178 of 198  vol = 834.473
core size 177 of 198  vol = 785.273
core size 176 of 198  vol = 762.127
core size 175 of 198  vol = 722.033
core size 174 of 198  vol = 700.399
core size 173 of 198  vol = 677.261
core size 172 of 198  vol = 657.813
core size 171 of 198  vol = 632.911
core size 170 of 198  vol = 614.201
core size 169 of 198  vol = 591.518
core size 168 of 198  vol = 573.982
core size 167 of 198  vol = 552.405
core size 166 of 198  vol = 529.491
core size 165 of 198  vol = 500.547
core size 164 of 198  vol = 482.52
core size 163 of 198  vol = 458.428
core size 162 of 198  vol = 444.457
core size 161 of 198  vol = 433.583
core size 160 of 198  vol = 419.088
core size 159 of 198  vol = 404.935
core size 158 of 198  vol = 393.804
core size 157 of 198  vol = 383.004
core size 156 of 198  vol = 366.655
core size 155 of 198  vol = 352.027
core size 154 of 198  vol = 335.663
core size 153 of 198  vol = 319.398
core size 152 of 198  vol = 307.936
core size 151 of 198  vol = 296.818
core size 150 of 198  vol = 284.29
core size 149 of 198  vol = 273.46
core size 148 of 198  vol = 261.978
core size 147 of 198  vol = 249.601
core size 146 of 198  vol = 237.954
core size 145 of 198  vol = 226.1
core size 144 of 198  vol = 213.266
core size 143 of 198  vol = 200.215
core size 142 of 198  vol = 187.505
core size 141 of 198  vol = 177.525
core size 140 of 198  vol = 167.372
core size 139 of 198  vol = 160.875
core size 138 of 198  vol = 154.454
core size 137 of 198  vol = 148.439
```

```
core size 136 of 198  vol = 142.13
core size 135 of 198  vol = 136.529
core size 134 of 198  vol = 130.769
core size 133 of 198  vol = 123.868
core size 132 of 198  vol = 117.609
core size 131 of 198  vol = 112.709
core size 130 of 198  vol = 106.361
core size 129 of 198  vol = 100.591
core size 128 of 198  vol = 95.718
core size 127 of 198  vol = 91.068
core size 126 of 198  vol = 86.862
core size 125 of 198  vol = 82.309
core size 124 of 198  vol = 78.554
core size 123 of 198  vol = 74.632
core size 122 of 198  vol = 70.489
core size 121 of 198  vol = 66.802
core size 120 of 198  vol = 62.901
core size 119 of 198  vol = 59.152
core size 118 of 198  vol = 55.75
core size 117 of 198  vol = 51.831
core size 116 of 198  vol = 48.3
core size 115 of 198  vol = 44.926
core size 114 of 198  vol = 42.417
core size 113 of 198  vol = 39.424
core size 112 of 198  vol = 37.381
core size 111 of 198  vol = 33.059
core size 110 of 198  vol = 28.153
core size 109 of 198  vol = 25.33
core size 108 of 198  vol = 22.509
core size 107 of 198  vol = 20.695
core size 106 of 198  vol = 18.754
core size 105 of 198  vol = 17.758
core size 104 of 198  vol = 16.712
core size 103 of 198  vol = 15.44
core size 102 of 198  vol = 14.745
core size 101 of 198  vol = 14.758
core size 100 of 198  vol = 13.11
core size 99 of 198  vol = 11.017
core size 98 of 198  vol = 8.967
core size 97 of 198  vol = 7.643
core size 96 of 198  vol = 6.326
core size 95 of 198  vol = 5.37
core size 94 of 198  vol = 4.312
```

```
 core size 93 of 198  vol = 3.391
 core size 92 of 198  vol = 2.697
 core size 91 of 198  vol = 1.911
 core size 90 of 198  vol = 1.577
 core size 89 of 198  vol = 1.144
 core size 88 of 198  vol = 0.826
 core size 87 of 198  vol = 0.594
 core size 86 of 198  vol = 0.494
 FINISHED: Min vol ( 0.5 ) reached
```

The identified core atom positions can now be used as a basis for a better superposition and written out to a directory `corefit_structures`

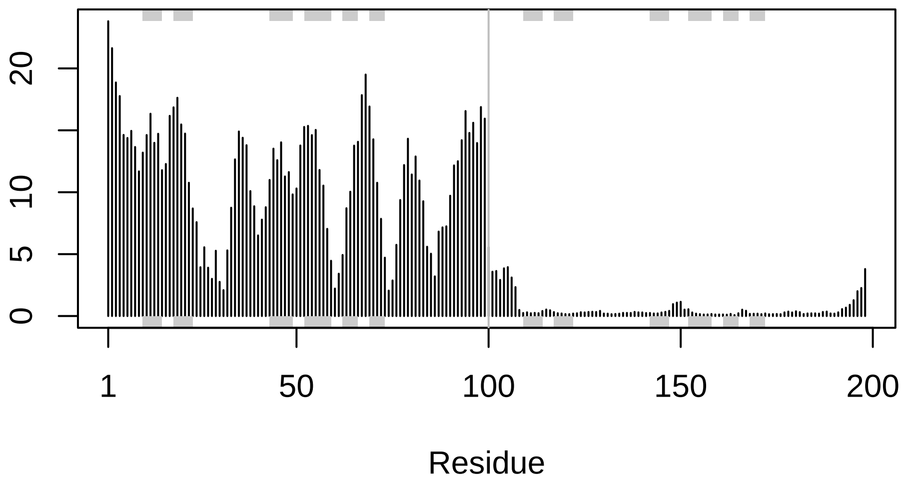
```
core.inds <- print(core, vol=0.5)
```

```
# 87 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     8  50     43
2    52  95     44
```

```
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")
```

The RMSF between positions of the structure can now be examined.

```
rf <- rmsf(xyz)

plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")
```



Predicted Alignment Error for Domains

Independent of the 3D structure, AlphaFold produces an output called Predicted Aligned Error (PAE). This is detailed in the JSON format result files, one for each model structure.

```
library(jsonlite)

# Listing of all PAE JSON files
results_dir <- "hivpr_dimer_23119"
pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                        full.names = TRUE)
```

```
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

attributes(pae1)
```

```
$names
[1] "plddt"   "max_pae" "pae"     "ptm"     "iptm"
```

```r
# Per-residue pLDDT scores
#  same as B-factor of PDB..
head(pae1$plddt)
```

```
[1] 91.44 93.75 94.19 93.44 95.62 89.81
```

The maximum PAE values are useful for ranking models. Here we can see that model 5 is much worse than model 1. The lower the PAE score the better. How about the other models, what are thir max PAE scores?

```r
pae1$max_pae
```

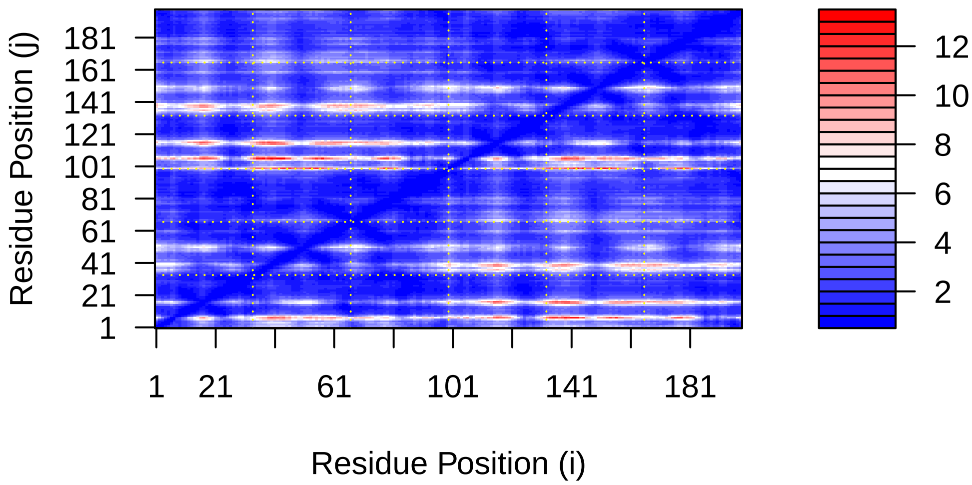
```
[1] 13.13281
```

```r
pae5$max_pae
```

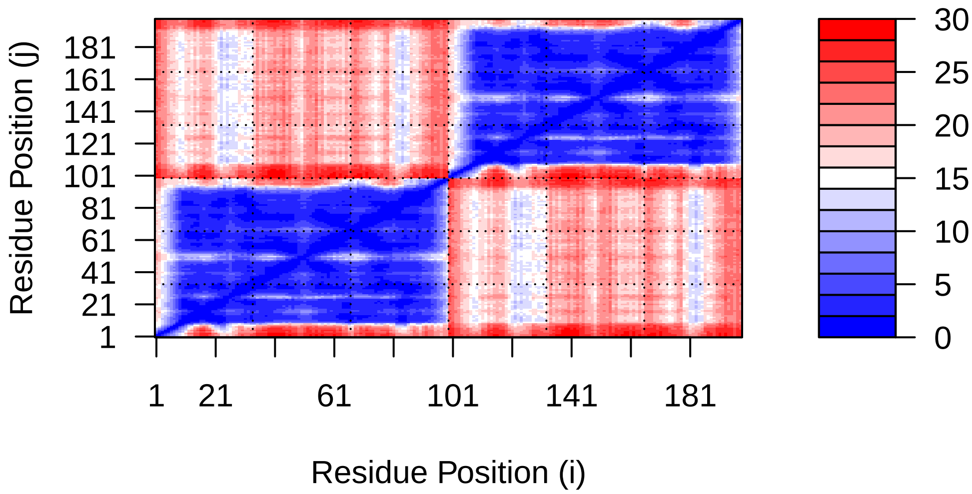
```
[1] 29.71875
```

We can plot the N by N (where N is the number of residues) PAE scores with ggplot or with functions from the Bio3D package:

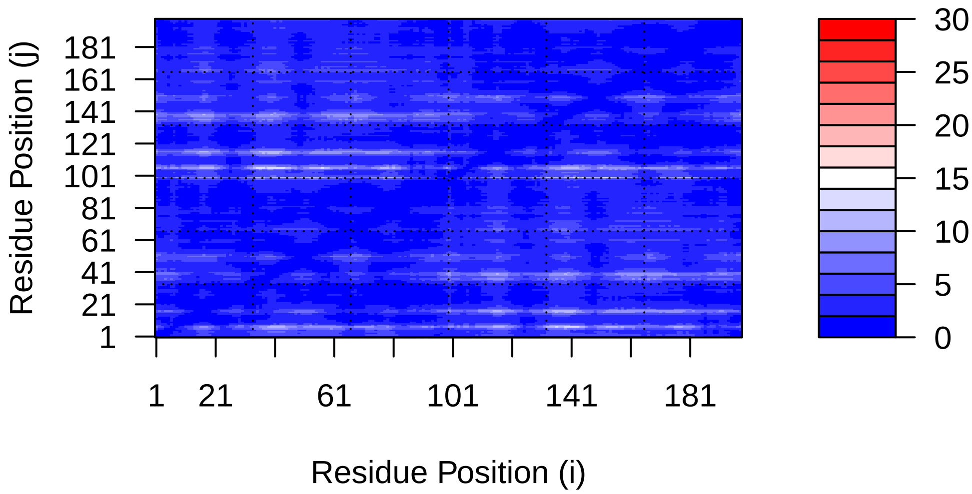
```r
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)")
```




```
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```




```r
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



Residue conservation from alignment file

```
aln_file <- list.files(path=results_dir,
                       pattern=".a3m$",
                        full.names = TRUE)
aln_file
```

```
[1] "hivpr_dimer_23119/hivpr_dimer_23119.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

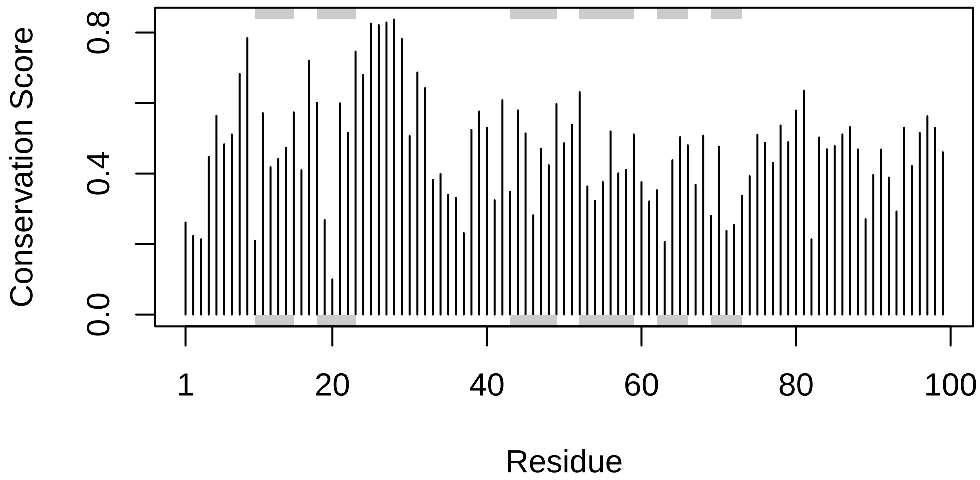
```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

```
dim(aln$ali)
```

```
[1] 5397  132
```

We can score residue conservation in the alignment with the `conserv()` function.

```
sim <- conserv(aln)
plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```




```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

For a final visualization of these functionally important sites we can map this conservation score to the Occupancy column of a PDB file for viewing in molecular viewer programs such as Mol*, PyMol, VMD, chimera etc.

```
m1.pdb <- read.pdb(pdbfiles[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```